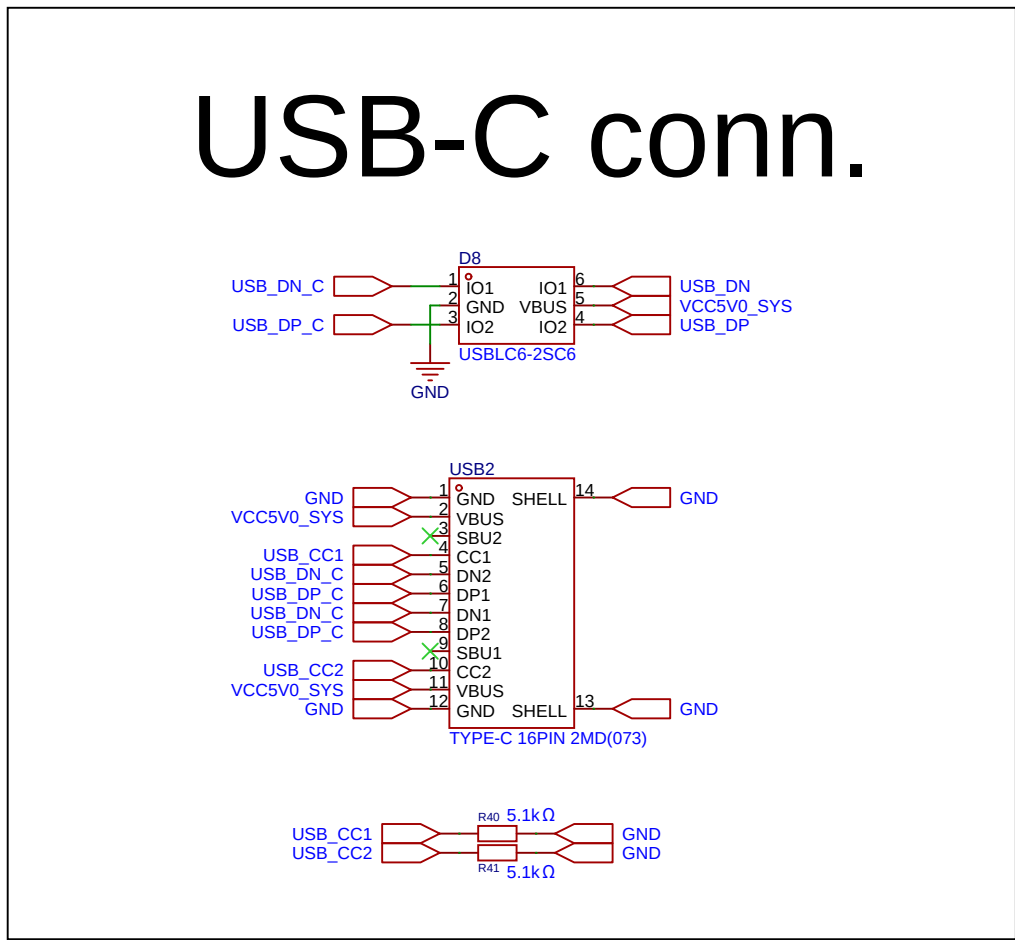
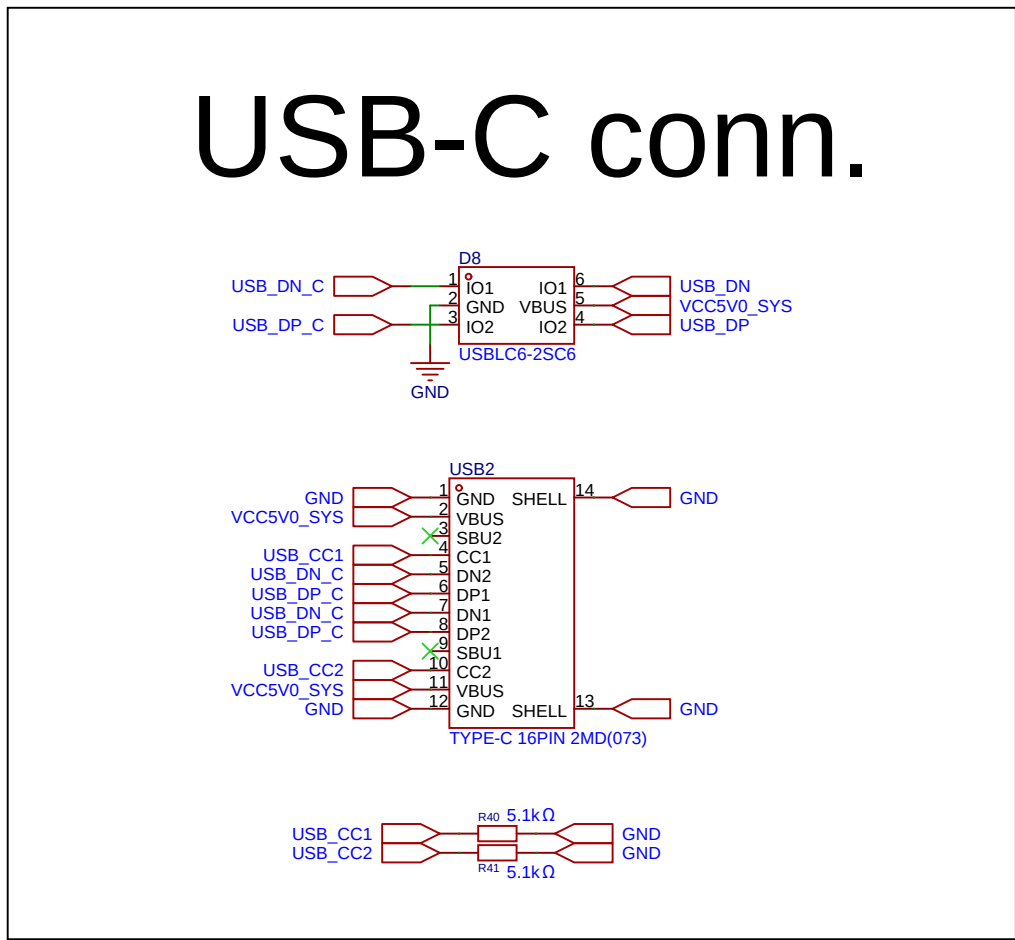


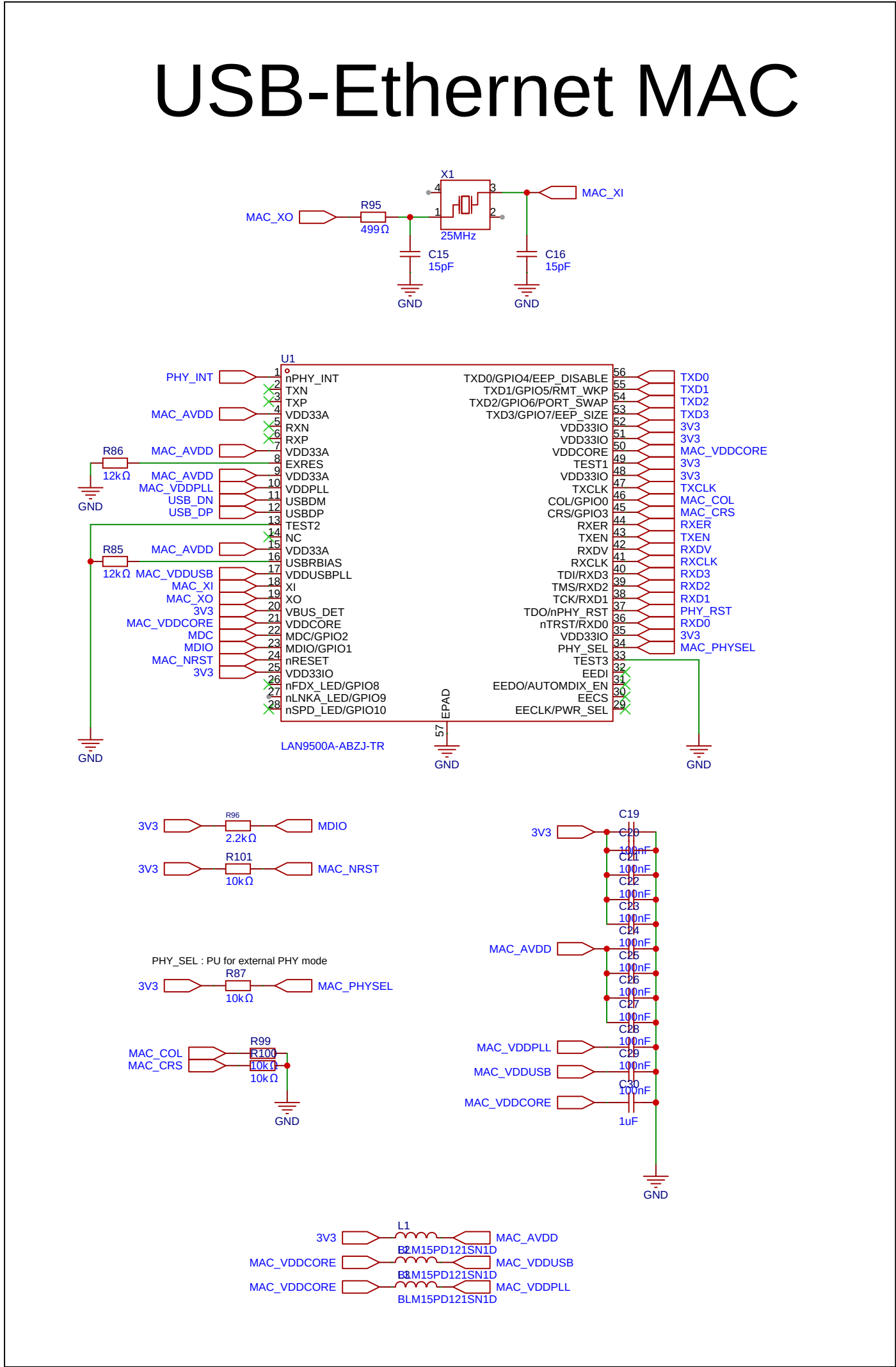
3V3 LDO + LED

[illegible]

USB-Ethernet MAC

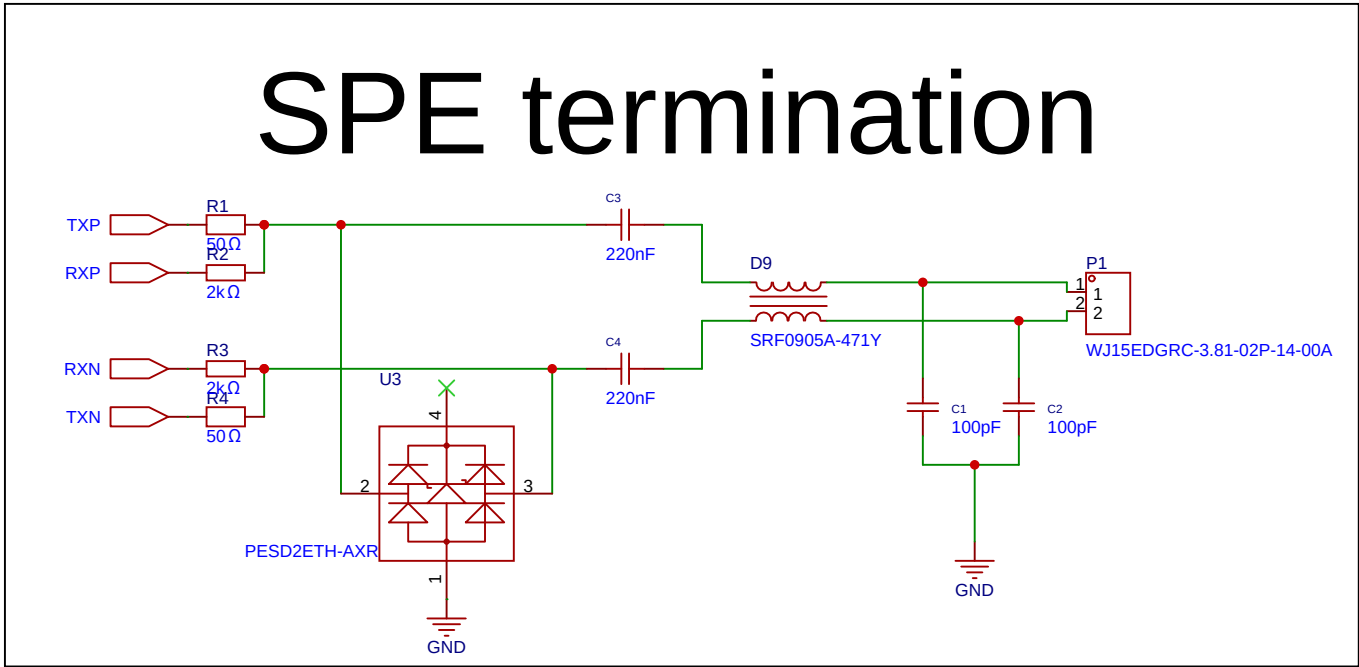
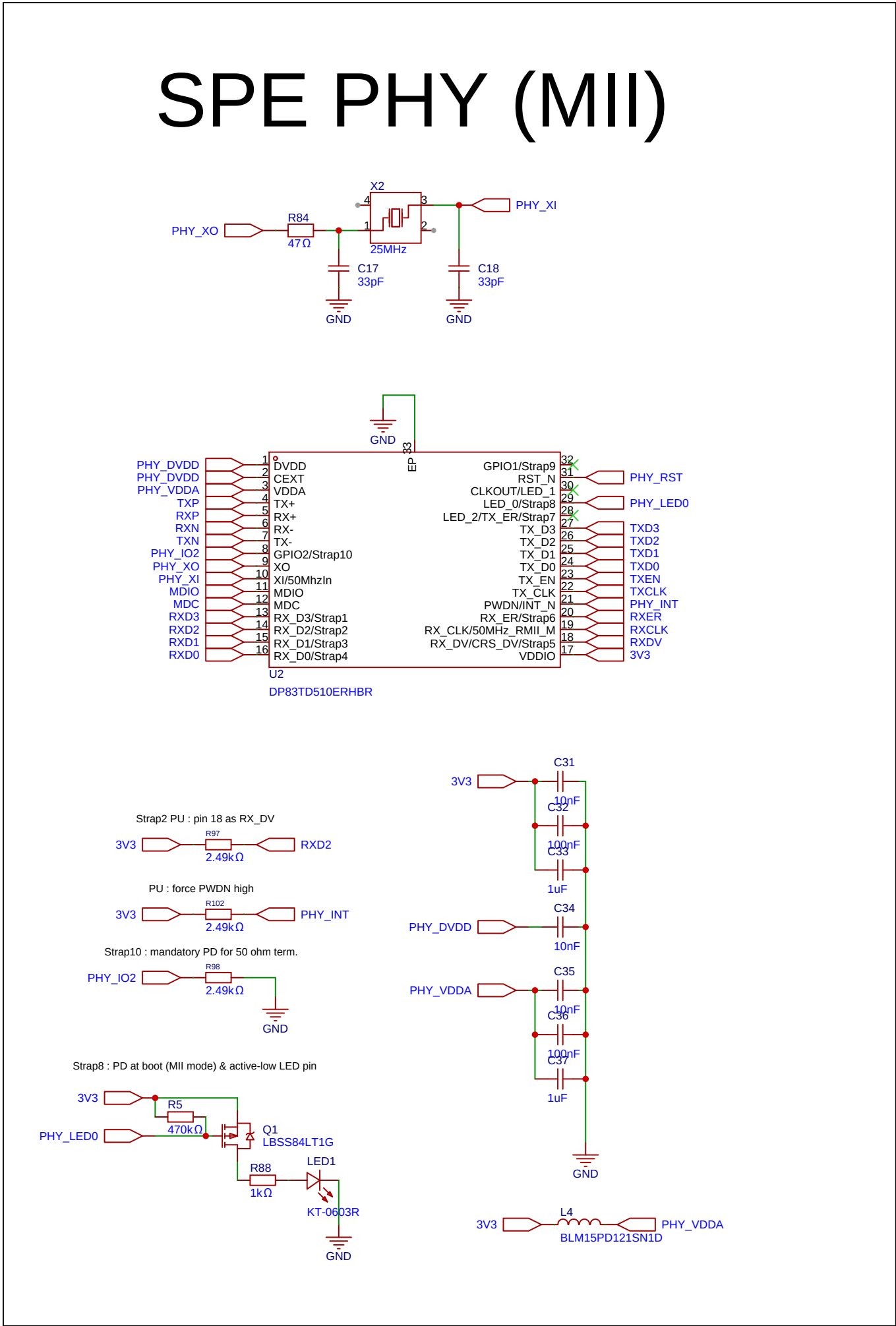
The schematic diagram illustrates the USB-Ethernet MAC circuit, centered around the LAN9500A-ABZJ-TR IC (U1). The circuit includes the following components and connections:

- Crystal and Clock:** A 25MHz crystal (X1) is connected to the MAC_XI pin. A 499Ω resistor (R95) is connected to the MAC_XO pin.
- Power Supply:** The IC is powered by 3V3 and GND. Key power pins include VDD33A, VDDPLL, VDDCORE, VDDUSB, and VDDPHYSEL.
- PHYSEL:** The PHYSEL pin is connected to the PHYSEL pin.
- Control Pins:** The IC has several control pins connected to 3V3 and GND, including MAC_COL, MAC_CRS, MAC_PHYSEL, MAC_VDDPLL, MAC_VDDUSB, and MAC_VDDCORE.
- USB-DP and USB-DN:** The IC is connected to a USB-DP pin (R85) and a USB-DN pin (R86).
- Other Pins:** The IC has various other pins connected to 3V3 and GND, including TXD0, TXD1, TXD2, TXD3, TXD4, TXD5, TXD6, TXD7, TXD8, TXD9, TXD10, TXD11, TXD12, TXD13, TXD14, TXD15, TXD16, TXD17, TXD18, TXD19, TXD20, TXD21, TXD22, TXD23, TXD24, TXD25, TXD26, TXD27, TXD28, TXD29, TXD30, TXD31, TXD32, TXD33, TXD34, TXD35, TXD36, TXD37, TXD38, TXD39, TXD40, TXD41, TXD42, TXD43, TXD44, TXD45, TXD46, TXD47, TXD48, TXD49, TXD50, TXD51, TXD52, TXD53, TXD54, TXD55, TXD56, TXD57, TXD58, TXD59, TXD60, TXD61, TXD62, TXD63, TXD64, TXD65, TXD66, TXD67, TXD68, TXD69, TXD70, TXD71, TXD72, TXD73, TXD74, TXD75, TXD76, TXD77, TXD78, TXD79, TXD80, TXD81, TXD82, TXD83, TXD84, TXD85, TXD86, TXD87, TXD88, TXD89, TXD90, TXD91, TXD92, TXD93, TXD94, TXD95, TXD96, TXD97, TXD98, TXD99, TXD100, TXD101, TXD102, TXD103, TXD104, TXD105, TXD106, TXD107, TXD108, TXD109, TXD110, TXD111, TXD112, TXD113, TXD114, TXD115, TXD116, TXD117, TXD118, TXD119, TXD120, TXD121, TXD122, TXD123, TXD124, TXD125, TXD126, TXD127, TXD128, TXD129, TXD130, TXD131, TXD132, TXD133, TXD134, TXD135, TXD136, TXD137, TXD138, TXD139, TXD140, TXD141, TXD142, TXD143, TXD144, TXD145, TXD146, TXD147, TXD148, TXD149, TXD150, TXD151, TXD152, TXD153, TXD154, TXD155, TXD156, TXD157, TXD158, TXD159, TXD160, TXD161, TXD162, TXD163, TXD164, TXD165, TXD166, TXD167, TXD168, TXD169, TXD170, TXD171, TXD172, TXD173, TXD174, TXD175, TXD176, TXD177, TXD178, TXD179, TXD180, TXD181, TXD182, TXD183, TXD184, TXD185, TXD186, TXD187, TXD188, TXD189, TXD190, TXD191, TXD192, TXD193, TXD194, TXD195, TXD196, TXD197, TXD198, TXD199, TXD200, TXD201, TXD202, TXD203, TXD204, TXD205, TXD206, TXD207, TXD208, TXD209, TXD210, TXD211, TXD212, TXD213, TXD214, TXD215, TXD216, TXD217, TXD218, TXD219, TXD220, TXD221, TXD222, TXD223, TXD224, TXD225, TXD226, TXD227, TXD228, TXD229, TXD230, TXD231, TXD232, TXD233, TXD234, TXD235, TXD236, TXD237, TXD238, TXD239, TXD240, TXD241, TXD242, TXD243, TXD244, TXD245, TXD246, TXD247, TXD248, TXD249, TXD250, TXD251, TXD252, TXD253, TXD254, TXD255, TXD256, TXD257, TXD258, TXD259, TXD260, TXD261, TXD262, TXD263, TXD264, TXD265, TXD266, TXD267, TXD268, TXD269, TXD270, TXD271, TXD272, TXD273, TXD274, TXD275, TXD276, TXD277, TXD278, TXD279, TXD280, TXD281, TXD282, TXD283, TXD284, TXD285, TXD286, TXD287, TXD288, TXD289, TXD290, TXD291, TXD292, TXD293, TXD294, TXD295, TXD296, TXD297, TXD298, TXD299, TXD300, TXD301, TXD302, TXD303, TXD304, TXD305, TXD306, TXD307, TXD308, TXD309, TXD310, TXD311, TXD312, TXD313, TXD314, TXD315, TXD316, TXD317, TXD318, TXD319, TXD320, TXD321, TXD322, TXD323, TXD324, TXD325, TXD326, TXD327, TXD328, TXD329, TXD330, TXD331, TXD332, TXD333, TXD334, TXD335, TXD336, TXD337, TXD338, TXD339, TXD340, TXD341, TXD342, TXD343, TXD344, TXD345, TXD346, TXD347, TXD348, TXD349, TXD350, TXD351, TXD352, TXD353, TXD354, TXD355, TXD356, TXD357, TXD358, TXD359, TXD360, TXD361, TXD362, TXD363, TXD364, TXD365, TXD366, TXD367, TXD368, TXD369, TXD370, TXD371, TXD372, TXD373, TXD374, TXD375, TXD376, TXD377, TXD378, TXD379, TXD380, TXD381, TXD382, TXD383, TXD384, TXD385, TXD386, TXD387, TXD388, TXD389, TXD390, TXD391, TXD392, TXD393, TXD394, TXD395, TXD396, TXD397, TXD398, TXD399, TXD400, TXD401, TXD402, TXD403, TXD404, TXD405, TXD406, TXD407, TXD408, TXD409, TXD410, TXD411, TXD412, TXD413, TXD414, TXD415, TXD416, TXD417, TXD418, TXD419, TXD420, TXD421, TXD422, TXD423, TXD424, TXD425, TXD426, TXD427, TXD428, TXD429, TXD430, TXD431, TXD432, TXD433, TXD434, TXD435, TXD436, TXD437, TXD438, TXD439, TXD440, TXD441, TXD442, TXD443, TXD444, TXD445, TXD446, TXD447, TXD448, TXD449, TXD450, TXD451, TXD452, TXD453, TXD454, TXD455, TXD456, TXD457, TXD458, TXD459, TXD460, TXD461, TXD462, TXD463, TXD464, TXD465, TXD466, TXD467, TXD468, TXD469, TXD470, TXD471, TXD472, TXD473, TXD474, TXD475, TXD476, TXD477, TXD478, TXD479, TXD480, TXD481, TXD482, TXD483, TXD484, TXD485, TXD486, TXD487, TXD488, TXD489, TXD490, TXD491, TXD492, TXD493, TXD494, TXD495, TXD496, TXD497, TXD498, TXD499, TXD500, TXD501, TXD502, TXD503, TXD504, TXD505, TXD506, TXD507, TXD508, TXD509, TXD510, TXD511, TXD512, TXD513, TXD514, TXD515, TXD516, TXD517, TXD518, TXD519, TXD520, TXD521, TXD522, TXD523, TXD524, TXD525, TXD526, TXD527, TXD528, TXD529, TXD530, TXD531, TXD532, TXD533, TXD534, TXD535, TXD536, TXD537, TXD538, TXD539, TXD540, TXD541, TXD542, TXD543, TXD544, TXD545, TXD546, TXD547, TXD548, TXD549, TXD550, TXD551, TXD552, TXD553, TXD554, TXD555, TXD556, TXD557, TXD558, TXD559, TXD560, TXD561, TXD562, TXD563, TXD564, TXD565, TXD566, TXD567, TXD568, TXD569, TXD570, TXD571, TXD572, TXD573, TXD574, TXD575, TXD576, TXD577, TXD578, TXD579, TXD580, TXD581, TXD582, TXD583, TXD584, TXD585, TXD586, TXD587, TXD588, TXD589, TXD590, TXD591, TXD592, TXD593, TXD594, TXD595, TXD596, TXD597, TXD598, TXD599, TXD600, TXD601, TXD602, TXD603, TXD604, TXD605, TXD606, TXD607, TXD608, TX



SPE termination

The schematic illustrates the SPE termination circuit. It features four input ports: TXP, RXP, RXN, and TXN. TXP and RXP are connected through resistors R1 (50Ω) and R2 (2kΩ) respectively. RXN and TXN are connected through resistors R3 (2kΩ) and R4 (50Ω) respectively. The signals from TXP and RXN are connected to the inputs of a quad differential line driver U3 (PESD2ETH-AXR). The outputs of U3 are connected to two capacitors, C3 (220nF) and C4 (220nF), which are connected to the primary terminals of a transformer D9 (SRF0905A-471Y). The secondary terminals of the transformer are connected to a connector P1 (WJ15EDGRC-3.81-02P-14-00A). The connector P1 is also connected to two capacitors, C1 (100pF) and C2 (100pF), which are connected to ground (GND).

[illegible]

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